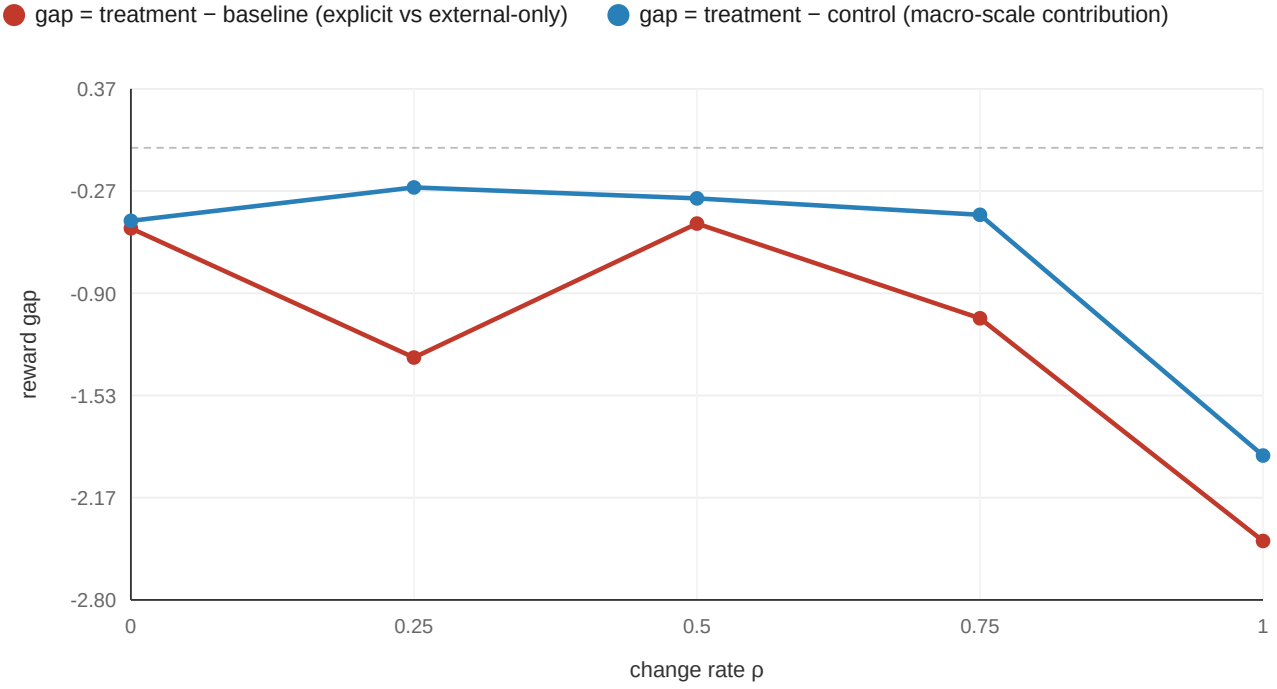
ρ-Sweep Report — Explicit-vs-Implicit Self-Observation under Non-Stationarity

Project #18 · QTI Continual Non-Stationary Navigation · Hypothesis H2b · PPO, 50,000 steps, 1 seed, 10 eval episodes

Hypothesis (H2b): H2b: The explicit-over-implicit advantage is near-zero at low rho and grows monotonically with rho, peaking under per-step change.

Main hypothesis: Given identical raw interaction history, an agent that explicitly computes multi-scale (micro/mezo/macro) self-observation features adapts better in non-stationary navigation than an external-state-only agent and a recurrent LSTM that integrates implicitly; the explicit-over-implicit advantage grows monotonically with the change rate rho.

Key result: explicit-vs-implicit gap vs change rate ρ



Gap-vs-ρ table

ρ	baseline (ext-only)	control (micro+mezo)	treatment (full)	gap (treat−base)	gap (treat−ctrl)
0	1.416	1.370	0.918	**-0.498**	**-0.452**
0.25	1.364	0.311	0.065	**-1.298**	**-0.245**
0.5	1.633	1.477	1.164	**-0.469**	**-0.313**
0.75	1.559	0.918	0.503	**-1.056**	**-0.415**
1	1.468	0.939	-0.966	**-2.434**	**-1.905**

Reward shaping, dynamics and termination are byte-identical across the three siblings, so observation content is the only manipulated variable; mean-reward gaps are therefore directly comparable.

Mean reward per variant (eval, deterministic)

variant	ρ=0	ρ=0.25	ρ=0.5	ρ=0.75	ρ=1
Baseline	1.416	1.364	1.633	1.559	1.468
Control	1.370	0.311	1.477	0.918	0.939
Treatment	0.918	0.065	1.164	0.503	-0.966

H2b verdict

H2b: PARTIAL / NON-MONOTONIC

non-monotonic in rho (endpoints gap(rho=0)=-0.498 -> gap(rho=1)=-2.434)

Notes & caveats

- This is a fast, uniform-budget look (50,000 steps, single seed) intended to populate the gap-vs-ρ curve immediately. A definitive 100k-step, 2-seed sweep is running in the background for the final paper.
- The "implicit" recurrent baseline (Recurrent PPO / PPO-LSTM, H2) is evaluated in the full sweep; this report's gap uses the external-state-only sibling as the explicit-vs-non-explicit reference.
- Total compute for this report: 1182 s.